

```
!pip install chromadb sentence-transformers pandas
```

```
Collecting chromadb
  Downloading chromadb-1.5.6-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (5.0 kB)
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.3.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Collecting build>=1.0.3 (from chromadb)
  Downloading build-1.4.2-py3-none-any.whl.metadata (5.8 kB)
Requirement already satisfied: pydantic>=2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.12.3)
Requirement already satisfied: pydantic-settings>=2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.13.1)
Collecting pybase64>=1.4.1 (from chromadb)
  Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl.metadata (8.7 kB)
Requirement already satisfied: uvicorn>=0.18.3 in /usr/local/lib/python3.12/dist-packages (from uvicorn[standard]>=0.18.3->chromadb) (0.42.0)
Requirement already satisfied: numpy>=1.22.5 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.0.2)
Requirement already satisfied: typing-extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.15.0)
Collecting onnxruntime>=1.14.1 (from chromadb)
  Downloading onnxruntime-1.24.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.2 kB)
Requirement already satisfied: opentelemetry-api>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.38.0)
Collecting opentelemetry-exporter-otlp-proto-grpc>=1.2.0 (from chromadb)
  Downloading opentelemetry_exporter_otlp_proto_grpc-1.40.0-py3-none-any.whl.metadata (2.6 kB)
Requirement already satisfied: opentelemetry-sdk>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.38.0)
Requirement already satisfied: tokenizers>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.22.2)
Collecting pypika>=0.48.9 (from chromadb)
  Downloading pypika-0.51.1-py2.py3-none-any.whl.metadata (51 kB)
52.0/52.0 kB 3.8 MB/s eta 0:00:00
Requirement already satisfied: tqdm>=4.65.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.67.3)
Requirement already satisfied: overrides>=7.3.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (7.7.0)
Requirement already satisfied: importlib-resources in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.5.2)
Requirement already satisfied: grpcio>=1.58.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.80.0)
Collecting bcrypt>=4.0.1 (from chromadb)
  Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl.metadata (10 kB)
Requirement already satisfied: typer>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.24.1)
Collecting kubernetes>=28.1.0 (from chromadb)
  Downloading kubernetes-35.0.0-py2.py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: tenacity>=8.2.3 in /usr/local/lib/python3.12/dist-packages (from chromadb) (9.1.4)
Requirement already satisfied: pyyaml>=6.0.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.0.3)
Requirement already satisfied: mmh3>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (5.2.1)
Requirement already satisfied: orjson>=3.9.12 in /usr/local/lib/python3.12/dist-packages (from chromadb) (3.11.8)
Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.28.1)
Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (13.9.4)
Requirement already satisfied: jsonschema>=4.19.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.26.0)
Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (5.0.0)
Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.8.0)
Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.10.0+cpu)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.6.1)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.16.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: packaging>=24.0 in /usr/local/lib/python3.12/dist-packages (from build>=1.0.3->chromadb) (26.0)
Collecting pyproject_hooks (from build>=1.0.3->chromadb)
  Downloading pyproject_hooks-1.2.0-py3-none-any.whl.metadata (1.3 kB)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (4.13.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (2026.2.25)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (3.11)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx>=0.27.0->chromadb) (0.16.0)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (3.25.2)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->sentence-transformers) (2025.3.0)
```

```
import chromadb
import pandas as pd
from sentence_transformers import SentenceTransformer
```

```
from google.colab import files
uploaded = files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving bbc-text.csv.zip to bbc-text.csv (1).zip

```
import zipfile

# Assuming the uploaded file is 'bbc-text.csv (1).zip'
zip_file_name = 'bbc-text.csv (1).zip'
extracted_file_name = 'bbc-text.csv' # The name of the csv file inside the zip

# Unzip the file
with zipfile.ZipFile(zip_file_name, 'r') as zip_ref:
    zip_ref.extractall()

# Now read the extracted CSV file
df = pd.read_csv(extracted_file_name)
df.head()
```

	category	text
0	tech	tv future in the hands of viewers with home th...
1	business	worldcom boss left books alone former worldc...
2	sport	tigers wary of farrell gamble leicester say ...
3	sport	yeading face newcastle in fa cup premiership s...
4	entertainment	ocean s twelve raids box office ocean s twelve...

```
print(df.columns)
print(df.shape)
```

```
Index(['category', 'text'], dtype='object')
(2225, 2)
```

```
df = df[['category', 'text']].dropna()
df = df.head(100) # you can increase later
df.head()
```

	category	text
0	tech	tv future in the hands of viewers with home th...
1	business	worldcom boss left books alone former worldc...
2	sport	tigers wary of farrell gamble leicester say ...
3	sport	yeading face newcastle in fa cup premiership s...
4	entertainment	ocean s twelve raids box office ocean s twelve...

```
client = chromadb.Client()
```

```
collection = client.create_collection(name="rag_collection")
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
```

To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in your Google Colab and resta
You will be able to reuse this secret in all of your notebooks.

Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
```

```
modules.json: 100% 349/349 [00:00<00:00, 30.0kB/s]
```

```
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 10.1kB/s]
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downl

```
README.md: 10.5k/? [00:00<00:00, 785kB/s]
```

```
sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 4.18kB/s]
```

```
config.json: 100% 612/612 [00:00<00:00, 50.4kB/s]
```

```
model.safetensors: 100% 90.9M/90.9M [00:00<00:00, 461MB/s]
```

```
Loading weights: 100% 103/103 [00:00<00:00, 441.81it/s, Materializing param=pooler.dense.weight]
```

```
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
```

```
Key | Status | |
-----+-----+--
embeddings.position_ids | UNEXPECTED | |
```

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

```
tokenizer_config.json: 100% 350/350 [00:00<00:00, 16.9kB/s]
```

```
vocab.txt: 232k/? [00:00<00:00, 8.69MB/s]
```

```
tokenizer.json: 466k/? [00:00<00:00, 13.8MB/s]
```

```
special_tokens_map.json: 100% 112/112 [00:00<00:00, 4.53kB/s]
```

```
config.json: 100% 190/190 [00:00<00:00, 18.2kB/s]
```

```
documents = df['text'].tolist()
ids = [f"doc{i+1}" for i in range(len(documents))]

print("Total Documents:", len(documents))
print("Sample ID:", ids[:5])
```

```
Total Documents: 100
Sample ID: ['doc1', 'doc2', 'doc3', 'doc4', 'doc5']
```

```
embeddings = model.encode(documents)
print("Embedding shape:", len(embeddings), len(embeddings[0]))
```

```
Embedding shape: 100 384
```

```
collection.add(
    documents=documents,
    embeddings=embeddings.tolist(),
    ids=ids
)
```

```
query = "What is artificial intelligence?"

query_embedding = model.encode([query])

results = collection.query(
    query_embeddings=query_embedding.tolist(),
    n_results=2
)

print(results)
```

```
{'ids': [['doc95', 'doc81']], 'embeddings': None, 'documents': [['amnesty chief laments war failure the lack of public outrage about the war on terror is a powerful ind
```

```
for doc in results['documents'][0]:
    print("\nRetrieved Document:\n")
    print(doc)
    print("-" * 80)
```

```
Retrieved Document:
```

```
amnesty chief laments war failure the lack of public outrage about the war on terror is a powerful indictment of the failure of human rights groups  amnesty internation
-----
```

```
Retrieved Document:
```

```
us company admits benin bribery a us defence and telecommunications company has agreed to pay $28.5m after admitting bribery in the west african state of benin. the ti
-----
```

```
def retrieve(query):
    query_embedding = model.encode([query])
    results = collection.query(
        query_embeddings=query_embedding.tolist(),
```

```

        n_results=2
    )
    return results['documents'][0]

```

```
print(retrieve("Explain machine learning"))
```

```
['microsoft gets the blogging bug software giant microsoft is taking the plunge into the world of blogging. it is launching a test service to allow people to publish b
```

```
print("\nQuery 1 Results:")
print(retrieve("Tell me about technology"))
```

```
print("\nQuery 2 Results:")
print(retrieve("What is politics?"))
```

```
print("\nQuery 3 Results:")
print(retrieve("Sports news"))
```

```
Query 1 Results:
```

```
['pandas benefit from wireless net the world s dwindling panda population is getting a helping hand from a wireless internet network. the wolong nature reserve in the
```

```
Query 2 Results:
```

```
['howard hits back at mongrel jibe michael howard has said a claim by peter hain that the tory leader is acting like an attack mongrel shows labour is rattled by th
```

```
Query 3 Results:
```

```
['news corp makes $5.4bn fox offer news corporation is seeking to buy out minority investors in fox entertainment group its broadcasting subsidiary for about $5.4bn (
```

```
metadatas = [{"category": cat} for cat in df['category'].tolist()]
```

```
client = chromadb.Client()
collection = client.create_collection(name="rag_collection_with_metadata")
```

```
collection.add(
    documents=documents,
    embeddings=embeddings.tolist(),
    ids=ids,
    metadatas=metadatas
)
```

```
query = "latest technology ideas"
query_embedding = model.encode([query])
```

```
results = collection.query(
    query_embeddings=query_embedding.tolist(),
    n_results=2
)
```

```
print(results)
```

```
{'ids': [['doc25', 'doc1']], 'embeddings': None, 'documents': [['mobile audio enters new dimension as mobile phones move closer to being a ubiquitous all-in-one media
```

```

for i in range(len(results['documents'][0])):
    print(f"\nResult {i+1}")
    print("Category:", results['metadatas'][0][i]['category'])
    print("Text:", results['documents'][0][i][:500])    # first 500 characters
    print("-" * 80)

```

Result 1

Category: tech

Text: mobile audio enters new dimension as mobile phones move closer to being a ubiquitous all-in-one media player audio is becoming ever more important. but how good

Result 2

Category: tech

Text: tv future in the hands of viewers with home theatre systems plasma high-definition tvs and digital video recorders moving into the living room the way people w

```

def retrieve(query, n_results=2):
    query_embedding = model.encode([query])
    results = collection.query(
        query_embeddings=query_embedding.tolist(),
        n_results=n_results
    )

    for i in range(len(results['documents'][0])):
        print(f"\nResult {i+1}")
        print("Category:", results['metadatas'][0][i]['category'])
        print("Text:", results['documents'][0][i][:500])
        print("-" * 80)

```

```
retrieve("machine learning and technology")
```

Result 1

Category: tech

Text: pandas benefit from wireless net the world s dwindling panda population is getting a helping hand from a wireless internet network. the wolong nature reserve in

Result 2

Category: tech

Text: tv future in the hands of viewers with home theatre systems plasma high-definition tvs and digital video recorders moving into the living room the way people w

```

df_ai = df[df['category'] == 'tech'].copy()
df_ai.head()

```

	category	text
0	tech	tv future in the hands of viewers with home th...
19	tech	games maker fights for survival one of britain...
20	tech	security warning over fbi virus the us feder...
21	tech	halo 2 heralds traffic explosion the growing p...
24	tech	mobile audio enters new dimension as mobile ph...

```
df.to_csv("processed_rag_dataset.csv", index=False)  
print("Saved successfully!")
```

Saved successfully!